Functional Canonical Correlation and Discriminant Analysis

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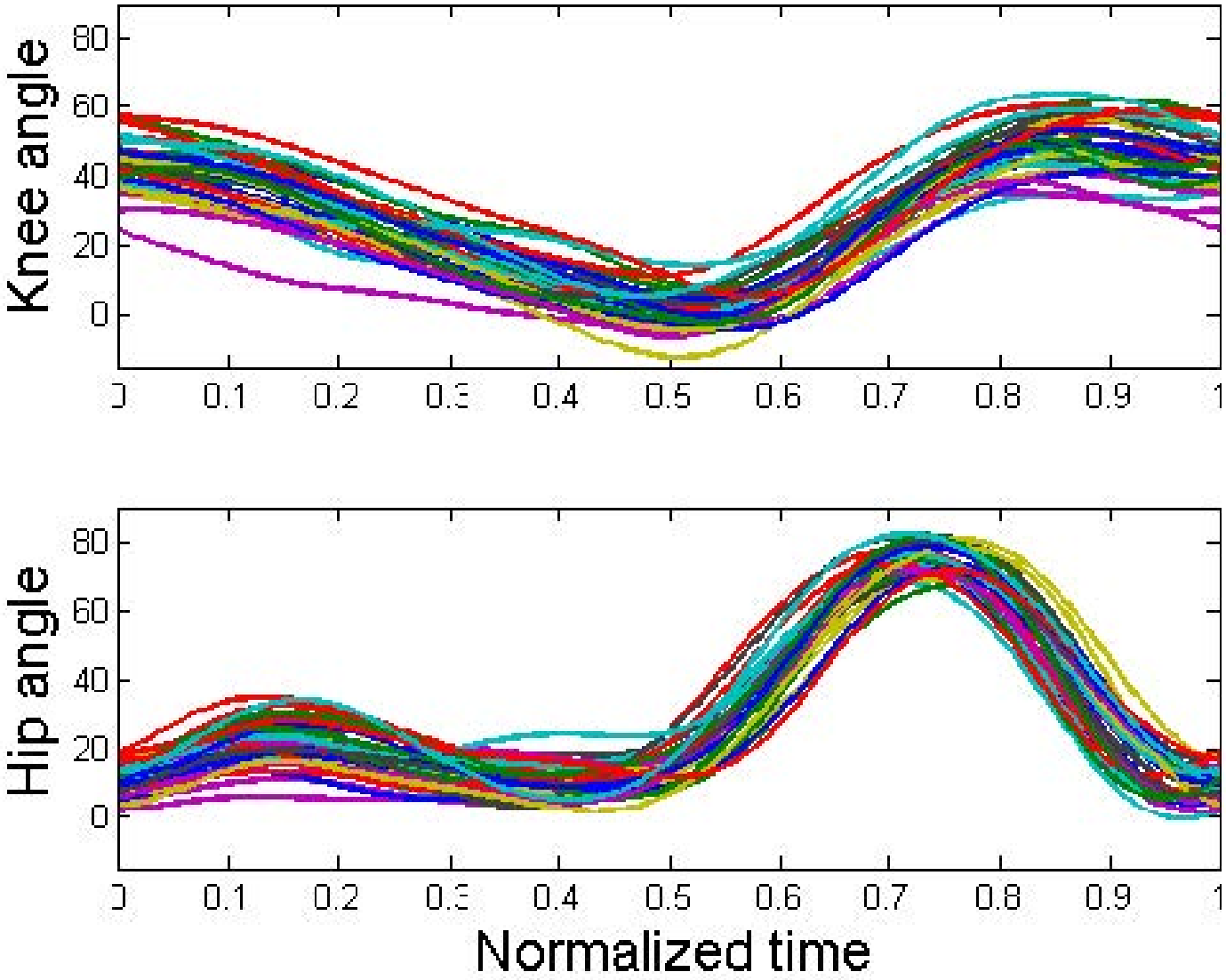
About canonical correlation analysis

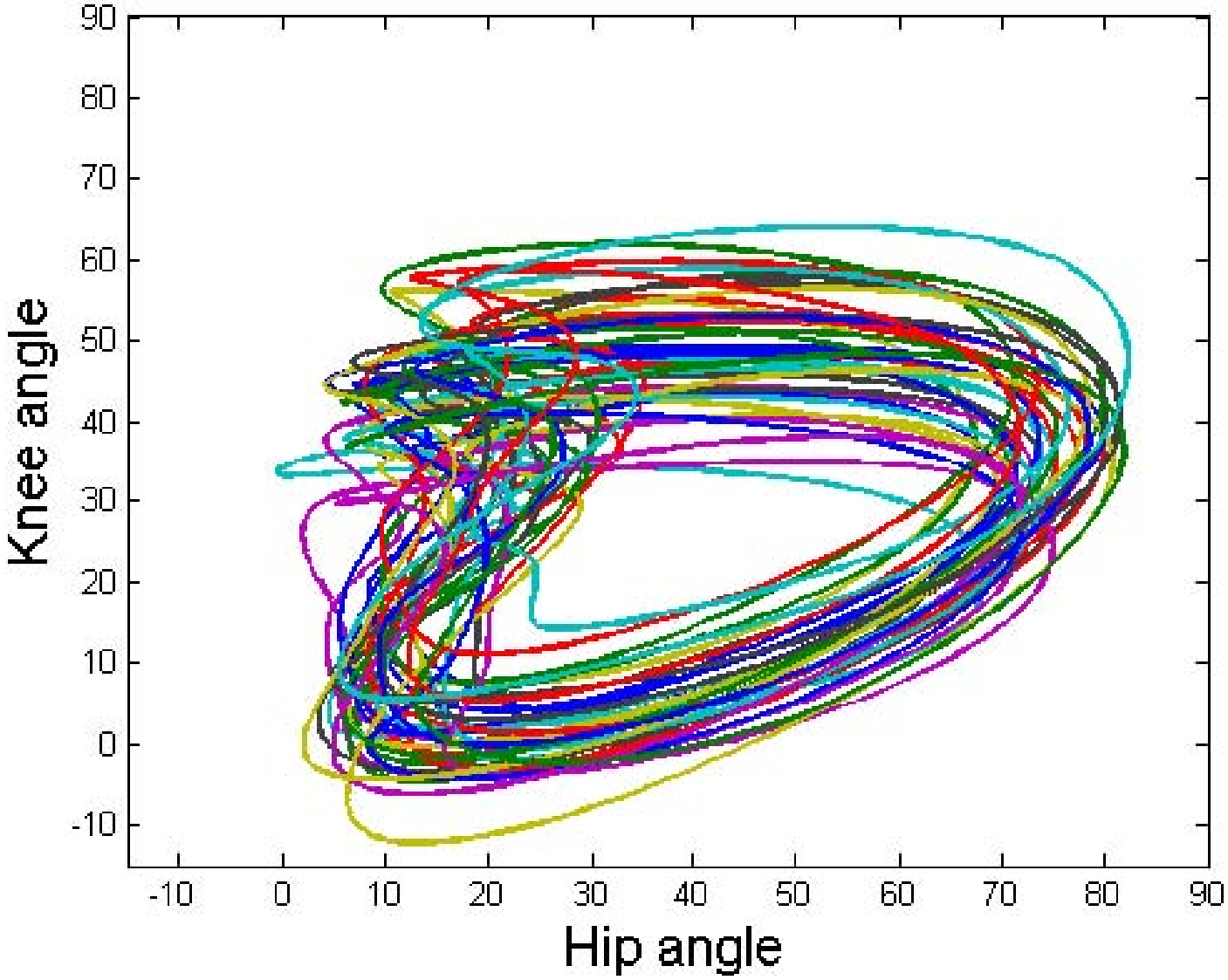
Consider the scenario of observing two sets of curves.

- PCA is usually used when we want to find the dominant *modes of variation* in the data.
- However, it does not address the issue of interaction between the two sets of curves.
- For this purpose, we want to look at the dominant *modes of correlation* between the two sets of functions.
- A tool in point is called *functional canonical correlation analysis*.

The hip-knee angle data



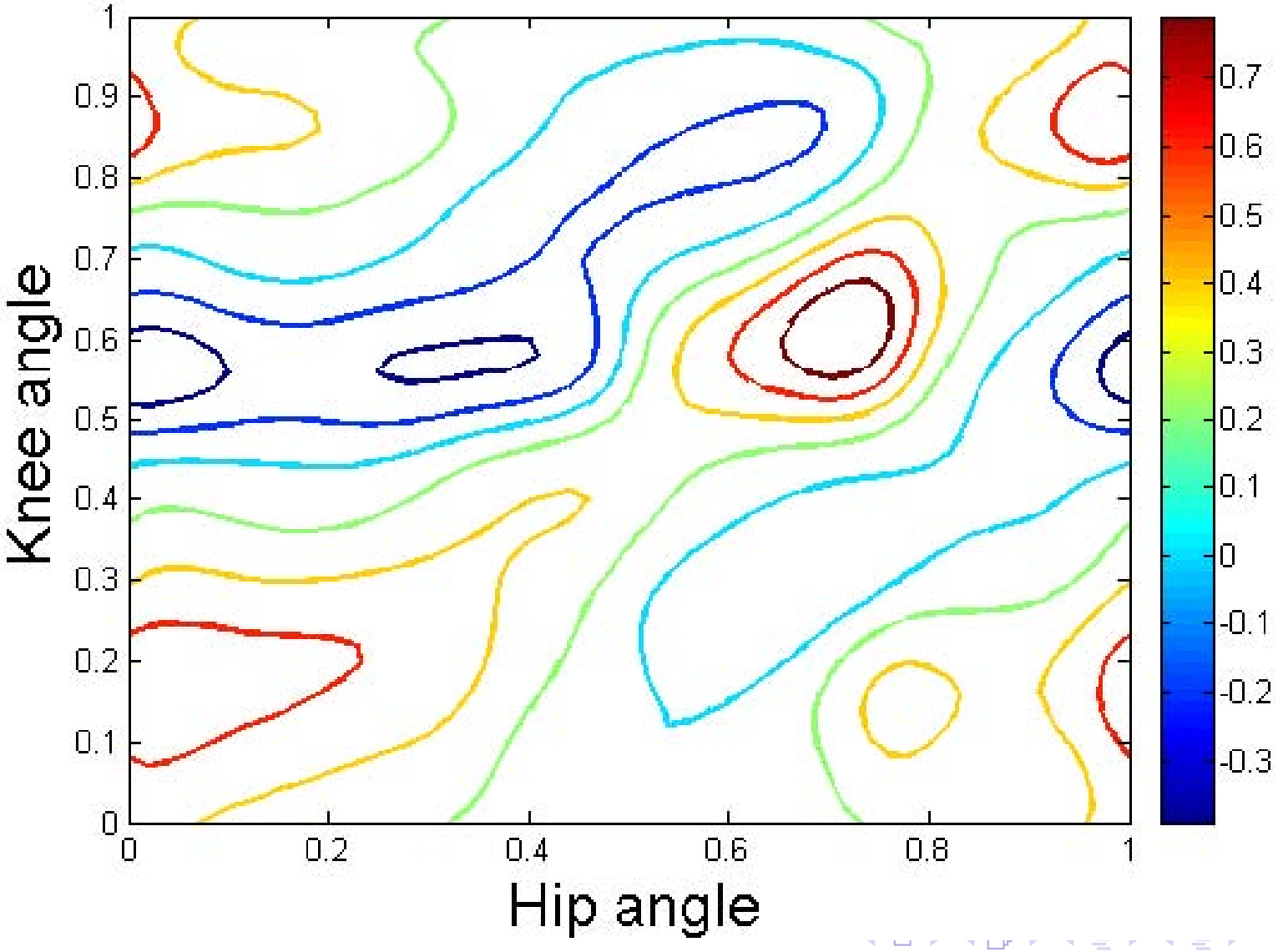
The hip-knee angles in 2D



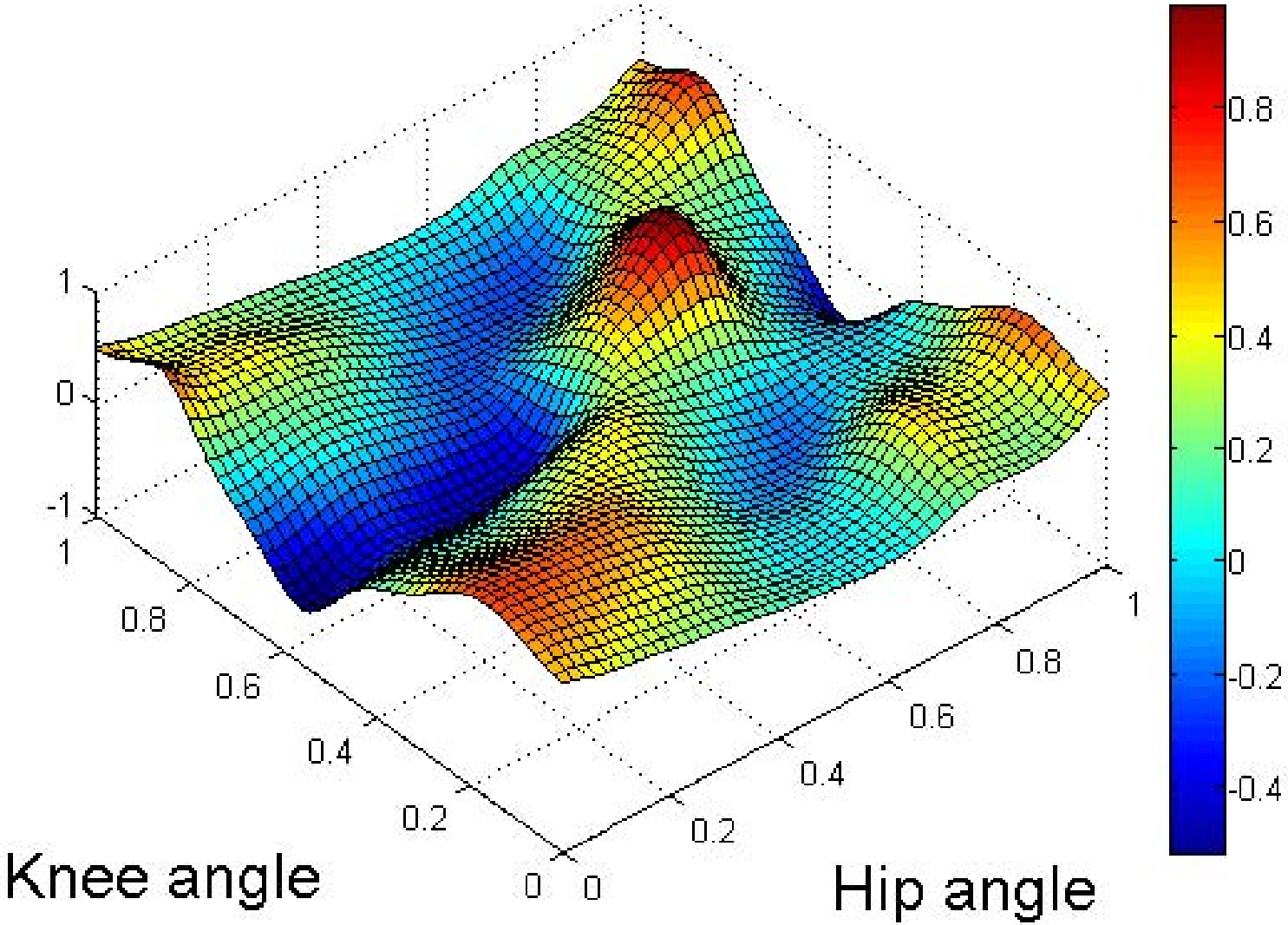
The hip-knee angle example

- We will explore the types of correlation between knee and hip angles for the gait data.
- We already have some preliminary correlation analysis results.

Contour plot of cross-correlations



Surface plot of cross-correlations



What do we see?

For the cross-correlation between hip angle Hip(t) and knee angle Knee(t), we see:

- positive cross-correlation on the main diagonal: hip angle Hip(t_1) and knee angle Knee(t_2) are mostly strongly correlated when t_1 and t_2 are approximately equal.
- The main ridge shows a slight reverse S shape: what does this mean?
- Differences in the behavior of the cross-correlation function at different points of the cycle: what does this mean?

Classical CCA: Step one

- Observing $(\mathbf{x}_i, \mathbf{y}_i), i = 1, \ldots, n$: $\mathbf{x}_i$ are p-vectors, $\mathbf{y}_i$ are q-vectors.
- Objective: Find *leading canonical variate weight vectors* $\boldsymbol{\xi}_1 = (\xi_{11}, ..., \xi_{p1})'$ and $\boldsymbol{\eta}_1 = (\eta_{11}, ..., \eta_{q1})'$ for which the *canonical variates* $\boldsymbol{\xi}_1'\mathbf{x}_i$ and $\boldsymbol{\eta}_1'\mathbf{y}_i$ are as highly correlated as possible.
- Further constraints:
  - $\{\boldsymbol{\xi}_1'\mathbf{x}_i\}$ and $\{\boldsymbol{\eta}_1'\mathbf{y}_i\}$ both **have sample variance equal to 1**,
  - the correlation ρ_1 between them is **positive**.

Classical CCA: Further steps

To find the jth subsidiary canonical variates,

- compute weight vectors $\boldsymbol{\xi}_j$ and $\boldsymbol{\eta}_j$ to maximize the sample correlation $\rho_j = \text{corr}(\boldsymbol{\xi}_j'\mathbf{x}_i, \boldsymbol{\eta}_j'\mathbf{y}_i)$ subject to the constraints that
  - $\text{corr}(\boldsymbol{\xi}_j'\mathbf{x}_i, \boldsymbol{\xi}_k'\mathbf{x}_i) = 0, k = 1, \ldots, j-1,$
  - $\text{corr}(\boldsymbol{\eta}_j'\mathbf{y}_i, \boldsymbol{\eta}_k'\mathbf{y}_i) = 0, k = 1, \ldots, j-1,$
  - $\text{corr}(\boldsymbol{\xi}_j'\mathbf{x}_i, \boldsymbol{\eta}_k'\mathbf{y}_i) = 0, k = 1, \ldots, j-1,$
  - $\text{corr}(\boldsymbol{\xi}_k'\mathbf{x}_i, \boldsymbol{\eta}_j'\mathbf{y}_i) = 0, k = 1, \ldots, j-1.$

Functional CCA: Probes for modes of correlation

- Two sets of functions: $x_i(t), y_i(t), i = 1, \ldots, N$.
- Define a *probe for the x functions*:

$$\int \xi(t) x_i(t) dt.$$

- Define a *probe for the y functions*:

$$\int \eta(t) y_i(t) dt.$$

- Given a pair of *probe weighting functions* (ξ, η), how strongly are the two sets of probe values correlated?

Functional CCA: Step one

- Re-phrase the question: What pair of probe weighting functions would give the highest possible correlation between probe values?
- The correlation that results is called the *first canonical correlation*, and indicated by ρ_1.
- Let the maximizing probe weighting functions be indicated by ξ_1 and η_1. These are called the *leading canonical weight functions*.
- The probe scores

$$f_{i1} = \int \xi_1(t)x_i(t)dt \text{ and } g_{i1} = \int \eta_1(t)y_i(t)dt$$

are called the *leading canonical variates*, or *leading canonical variables*.

Functional CCA: Further steps

- We can repeat this process. Let a new pair of canonical weight functions ξ_2 and η_2 be defined, and let them satisfy the two restrictions

$$\int \xi_1(t)\xi_2(t)dt = 0 \text{ and } \int \eta_1(t)\eta_2(t)dt = 0.$$

- We can again ask for the pair of ξ_2 and η_2 that maximize the correlation ρ_2 between the pairs of canonical variable values (f_{i2}, g_{i2}) defined by these canonical weight functions.
- And so on, until there are no more interesting canonical correlations to be found.

Functional CCA: Constraints and regularization

Consider the computation of the leading weight functions (ξ_1, η_1). To ensure the uniqueness and meaningfulness of (ξ_1, η_1),

- we need the following constraints on the sample variances of $\{f_{i1}, i = 1, \ldots, N\}$ and $\{g_{i1}, i = 1, \ldots, N\}$:

$$\mathrm{var}(f_{i1}) = \mathrm{var}(g_{i1}) = 1.$$

- We also need regularization such that the above constraints change to

$$\mathrm{var}(\int \xi x_i) + \lambda \|D^2 \xi\|^2 = \mathrm{var}(\int \eta y_i) + \lambda \|D^2 \eta\|^2 = 1.$$

Smoothed functional CCA

- Alternatively, compute (ξ_1, η_1) that maximize

$$\text{ccorsq}_\lambda(\xi, \eta) = \frac{\{\text{cov}(\int \xi x_i, \int \eta y_i)\}^2}{\{\text{var}(\int \xi x_i) + \lambda \|D^2 \xi\|^2\}\{\text{var}(\int \eta y_i) + \lambda \|D^2 \eta\|^2\}}.$$

  subject to the constraints $\int \xi_1^2 = \int \eta_1^2 = 1$.
- This procedure is referred to as *smoothed functional canonical correlation analysis*.

Why regularization?

- Classical CCA requires $n > p + q + 1$ to prevent the sample covariance matrix of the n vectors $(\mathbf{x}_i, \mathbf{y}_i)$ from being singular, which can produce a rather large number of large correlations that are often difficult to interpret.
- Functional CCA also needs regularization on the weight functions to avoid obtaining bumpy canonical weight functions, which tend to emphasize higher frequency modes of variation that are usually impossible to interpret, or else are not very interesting.
- More details in Leurgans, Moyeed and Silverman (1993).

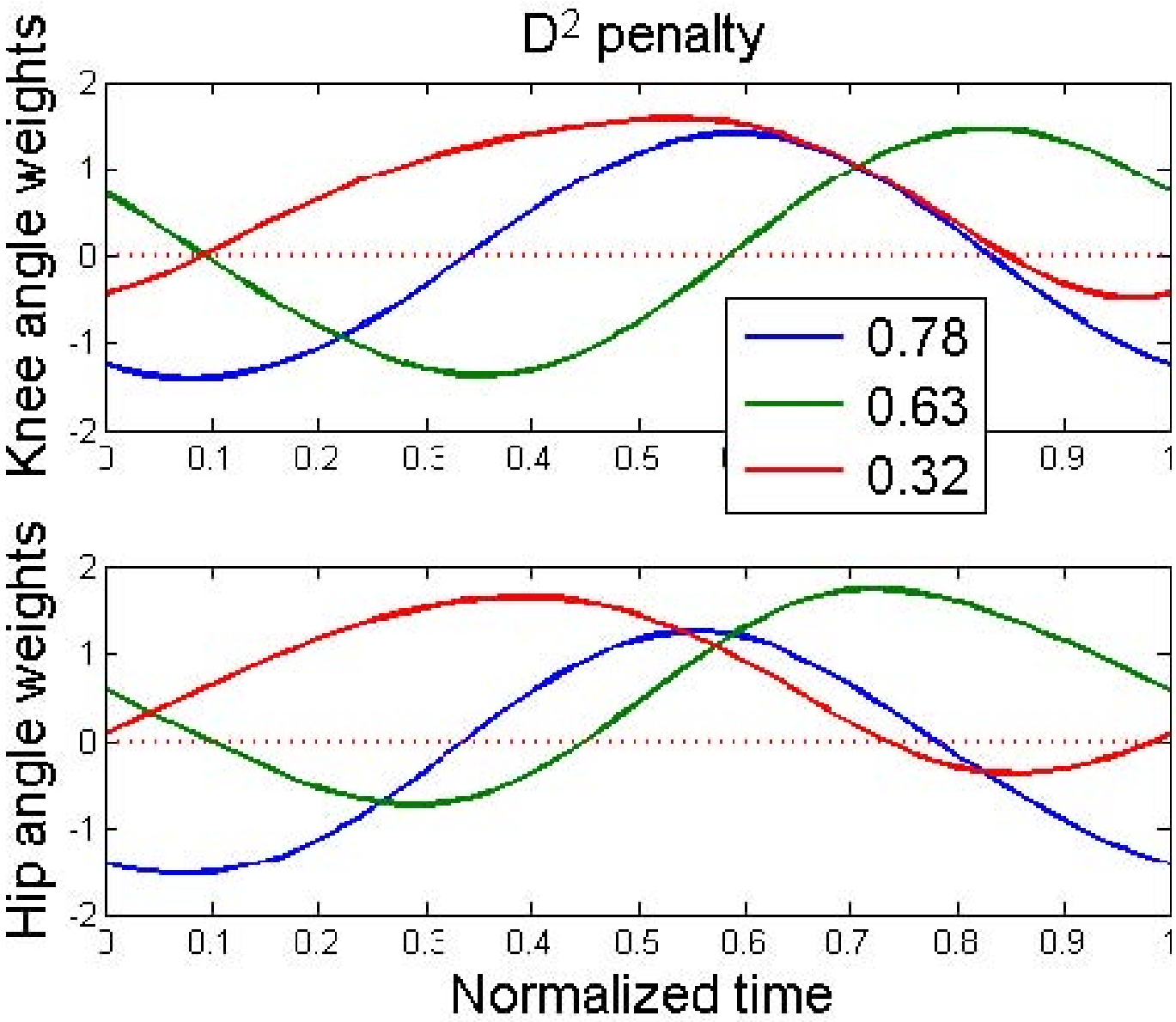
Choice of smoothing parameter

- For the leading canonical variates, λ is chosen by a leave-one-out cross-validation criterion:
  - let $\text{ccorsq}_\lambda^{(-i)}(\xi, \eta)$ be the sample penalized squared correlation calculated with the observation (x_i, y_i) removed.
  - let $(\xi_\lambda^{(-i)}, \eta_\lambda^{(-i)})$ be the functions that maximize $\text{ccorsq}_\lambda^{(-i)}(\xi, \eta)$.
  - the cross-validation score for λ is defined as the squared correlation of the N pairs of numbers $(\int \xi_\lambda^{(-i)} x_i, \int \eta_\lambda^{(-i)} y_i)$.
  - choose λ to maximize this cross-validation score.
- Subsequent canonical variates use the same λ.
- Subjective choice of smoothing parameters is acceptable.

Functional CCA: Gait data example

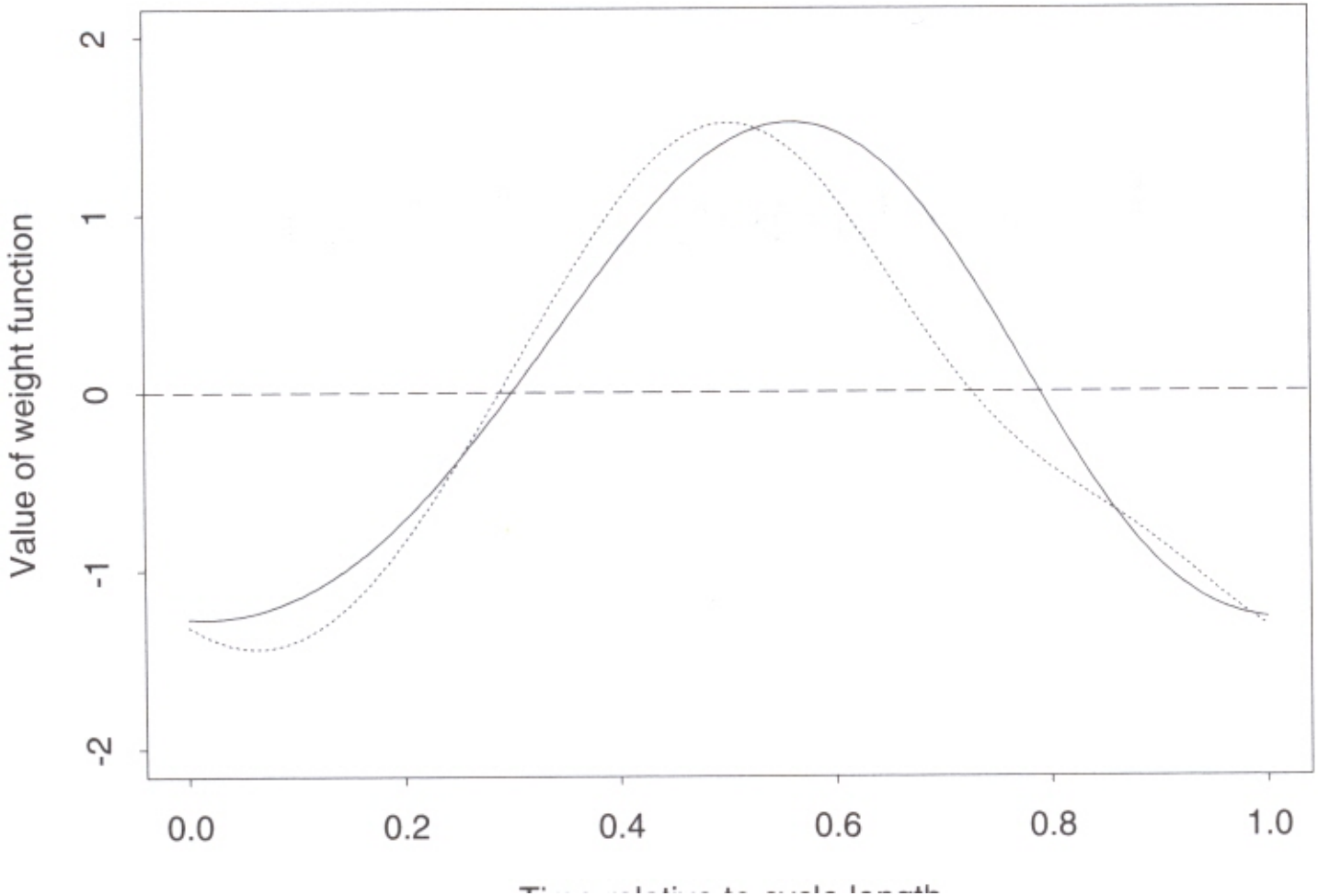
- In order to explore the correlations between knee and hip angles for the gait data, we first used D^2 to define both penalties on the roughness of each pair of weighting functions.
- The weights placed on these two penalties were $\lambda = 0.001$.
- The first three canonical correlations were 0.73, 0.58 and 0.31. The next was 0.05.
- It seemed that there were only three important modes of correlation between knee and hip angle.

FCCA for Gait data: D^2 penalty



FCCA for Gait data: 1st weigth

Solid: hip weight. Dotted: knee weight.

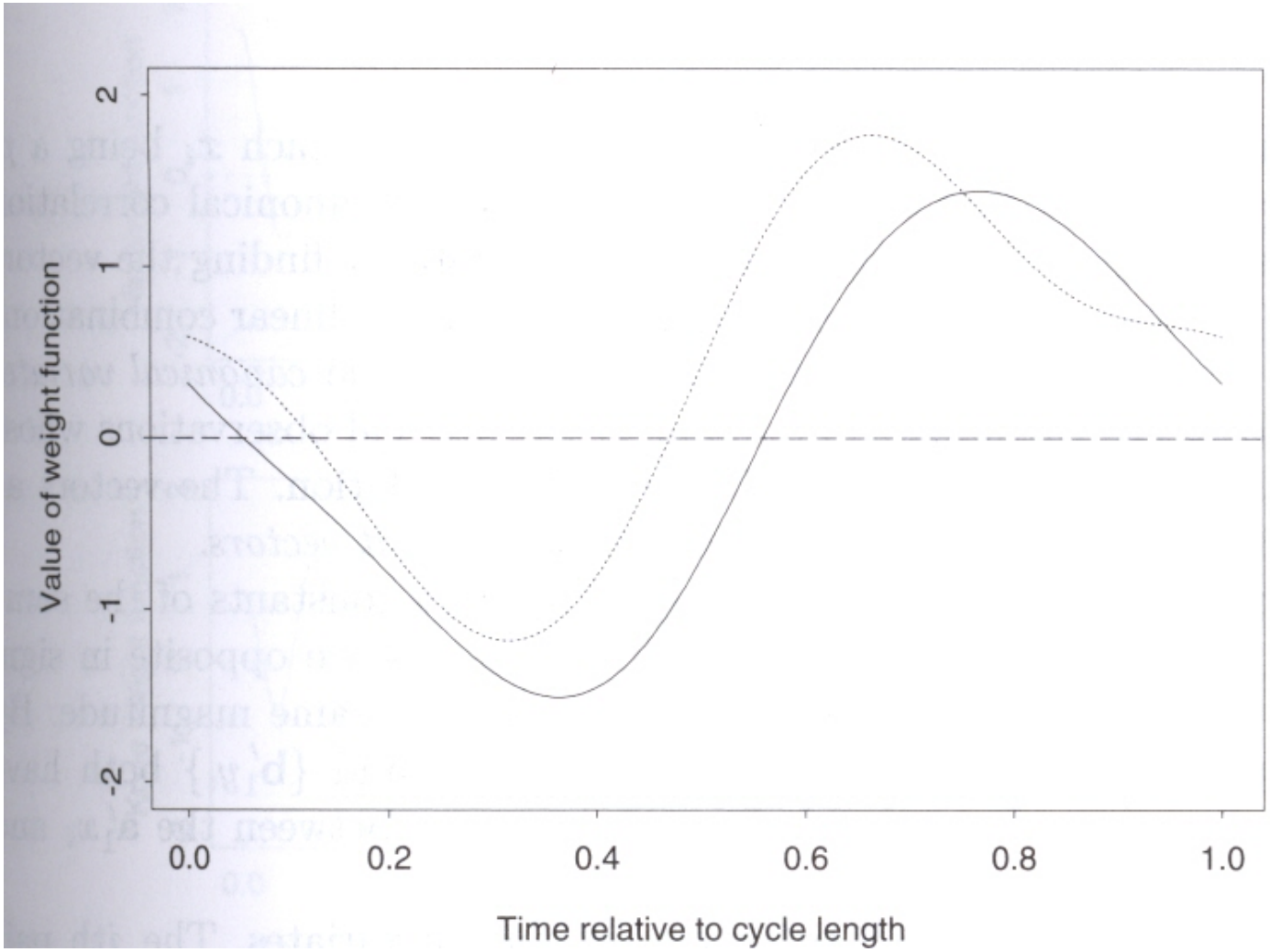


What do we see?

- Similar shapes for ξ_1 and η_1: there is correlation between the two angles Hip(t) and Knee(t) at any time.
- In the middle of the cycle, the extreme in the hip weight curve follows that in the knee weight curve.
- Near the ends of the cycle, the effects occur in the opposite order.
- Propagation of the heel-strike effect again?

FCCA for Gait data: 2nd weigth

Solid: hip weight. Dotted: knee weight.



What do we see?

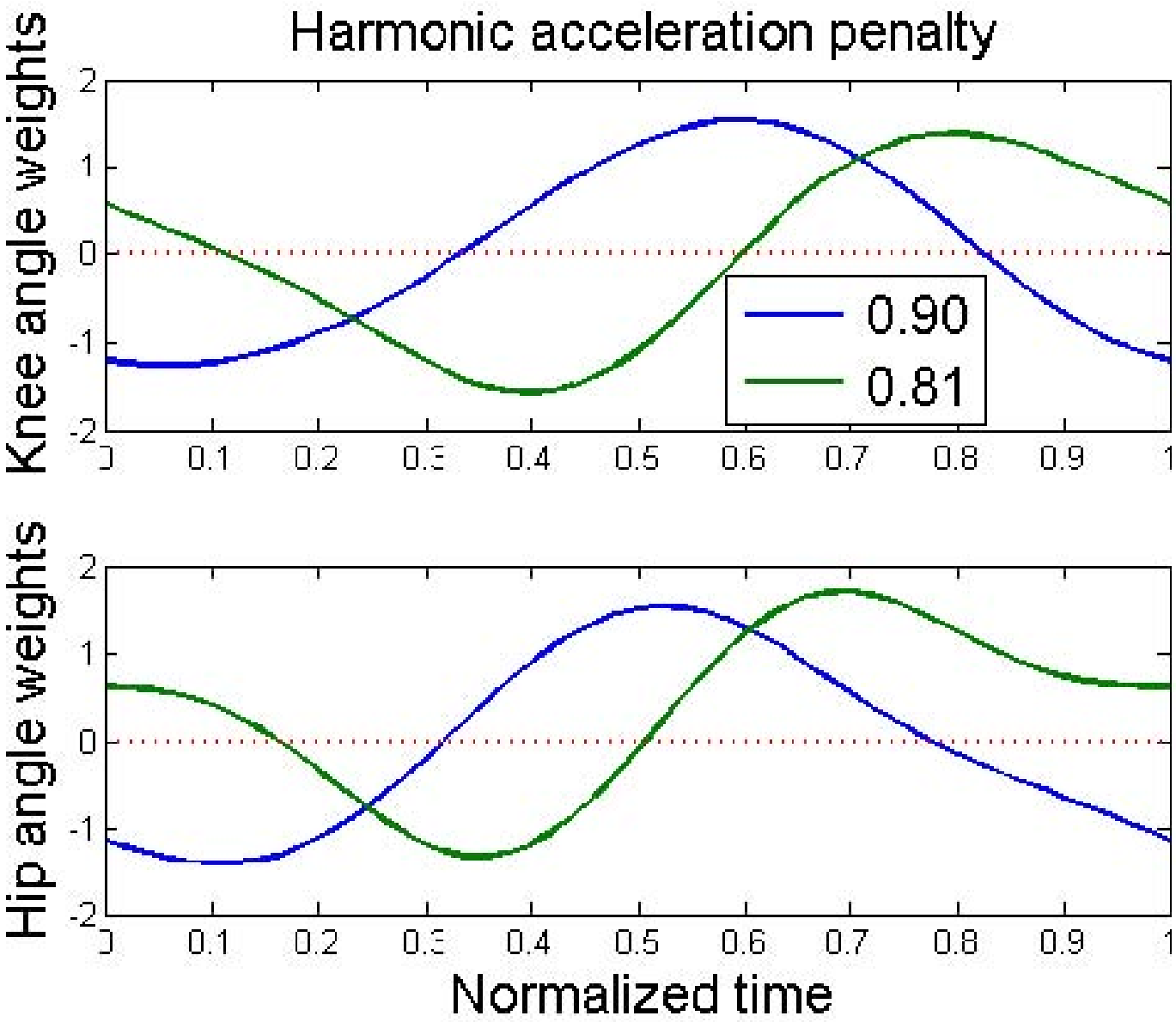
- In the middle of the cycle, the hip weight curve crosses zero considerably later than the knee curve.
- Near the begging of the cycle, the hip weight curve crosses first.

For each of the two leading pairs of weight functions, we could roughly transform them to be identical for the hip and the knee by speeding up the hip cycle relative the knee cycle in the first half of the cycle, and slowing it down in the second.

Gait data: harmonic penalty

- But since the functions are essentially periodic, we decided that the harmonic acceleration operator would be a better choice of penalties.
- Using $\lambda = 10^{-6}$, we found the first four canonical correlations to be 0.90, 0.81, 0.39 and 0.26.
- This analysis highlighted only two dominant modes of correlation.

FCCA for Gait data: harmonic penalty



What do we see?

- Similar to what we see using the D^2 penalty.
- The shift effect stands out more clearly now.

Basis expansion

- Suppose $\boldsymbol{\phi}(t) = (\phi_1, \ldots, \phi_M)$ is a vector of M basis functions.
- Let $\mathbf{K}$ be the matrix with entries $\int (D^2\phi_j)(D^2\phi_k)$ and $\mathbf{J}$ be the matrix with entries $\int \phi_j \phi_k$
- Let $\mathbf{C}$ and $\mathbf{D}$ be $N \times K$ coefficient matrices such that

$$\mathbf{x}(t) = \mathbf{C}\boldsymbol{\phi}(t) \text{ and } \mathbf{y}(t) = \mathbf{D}\boldsymbol{\phi}(t)$$

- and the expansions for weight functions be

$$\xi_j(s) = \mathbf{a}_j'\boldsymbol{\phi}(s) \text{ and } \eta_j(s) = \mathbf{b}_j'\boldsymbol{\phi}(s).$$

Generalized eigenvalue equation

- Define $M \times M$ covariance matrices $\tilde{\mathbf{V}}_{11}, \tilde{\mathbf{V}}_{12}, \tilde{\mathbf{V}}_{22}$ to be the matrices with (ν, μ) entries

$$N^{-1} \sum_{i=1}^{N} c_{i\nu} c_{i\mu}, \; N^{-1} \sum_{i=1}^{N} c_{i\nu} d_{i\mu}, \; N^{-1} \sum_{i=1}^{N} d_{i\nu} d_{i\mu},$$

- It can be shown that the smoothed FCCA boils down to solving the following generalized eigenvalue problem:

$$\begin{bmatrix} \mathbf{0} & \mathbf{J}\tilde{\mathbf{V}}_{12}\mathbf{J} \\ \mathbf{J}\tilde{\mathbf{V}}_{21}\mathbf{J} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{a}_j \\ \mathbf{b}_j \end{bmatrix} = \rho \begin{bmatrix} \mathbf{J}\tilde{\mathbf{V}}_{11}\mathbf{J} + \lambda \mathbf{K} & \mathbf{0} \\ \mathbf{0} & \mathbf{J}\tilde{\mathbf{V}}_{22}\mathbf{J} + \lambda \mathbf{K} \end{bmatrix} \begin{bmatrix} \mathbf{a}_j \\ \mathbf{b}_j \end{bmatrix}.$$

Reference paper

Hastie, T., Buja, A. and Tibshirani, R. (1995) Penalized discriminant analysis. *Annals of Statistics*, **23**, 73-102.

The optimal score problem

- Observations: $(x_i, y_i), i = 1, \ldots, N$, where each x_i is a function and each y_i is a class label from $\{1, \ldots, J\}$.
- Rewrite y_i as a J-vector $\mathbf{y}_i$: $\mathbf{y}_i$ has value 1 in position j and 0 elsewhere, if $y_i = j$.
- Look for a function β and a J-vector $\boldsymbol{\theta}$ that minimize the criterion

$$\mathrm{OSERR}(\boldsymbol{\theta}, \beta) = N^{-1} \sum_{i=1}^{N} \left[\int \beta x_i - \boldsymbol{\theta}' \mathbf{y}_i \right]^2$$

subject to the normalization constraint $N^{-1} \sum_i (\boldsymbol{\theta}' \mathbf{y}_i)^2 = 1$.

The optimal score problem: Main idea

- θ_j: score for category j.
- The criterion

$$\text{OSERR}(\boldsymbol{\theta}, \beta) = N^{-1} \sum_{i=1}^{N} \left[\int \beta x_i - \boldsymbol{\theta}' \mathbf{y}_i \right]^2$$

chooses the *optimal scores* θ_j to give the best available prediction from a linear property $\int \beta x$ of the observed functional data.

The optimal score problem: Regularization

- Given $\boldsymbol{\theta}$, the problem of finding β is finding a function which satifies a finite number of linear constraints.
- Too many candidates including one that gives perfect prediction of any specified values $\boldsymbol{\theta}'\mathbf{y}_i$.
- Hastie et al. (1995) introduced the *penalized* optimal scoring criterion

$$\mathrm{OSERR}_\lambda(\boldsymbol{\theta}, \beta) = \mathrm{OSERR}(\boldsymbol{\theta}, \beta) + \lambda \mathrm{PEN}(\beta).$$

The discriminant problem

- Observations: $x_i, i = 1, \ldots, N$, where each x_i is a function allocated to a category in $\{1, \ldots, J\}$.
- For any proposed linear discriminant functional $\int \beta x_i$, define

$$\theta_j = \text{average}(\{\int \beta x_i, x_i \text{ in category } j\})$$

.

- Hence, for any fixed β, the J-vector $\boldsymbol{\theta}$ minimizes the criterion OSERR$(\boldsymbol{\theta}, \beta)$, which can be interpreted as the *within-class variance* of the $\int \beta x_i$.
- The *between-class variance* is the variance of the discriminant class means $\boldsymbol{\theta}' \mathbf{y}_i$.

Discriminant analysis

- Objective: maximize the between-class variance subject to a constraint on the within-class variance.
- optimal scoring $\rightarrow$ discriminant analysis: role exchanges of the objective function and constraint; shift of focus from the score vector $\boldsymbol{\theta}$ to the discriminant functional defined by the function β.
- *Penalized discriminant analysis* maximizes the raw between-class variance subject to a penalized constraint on the within-class variance $\boldsymbol{\theta}'\mathbf{y}_i$.

$$\mathrm{OSERR}(\boldsymbol{\theta}, \beta) + \lambda \mathrm{PEN}(\beta) = 1.$$

Relationship with CCA

- Both the optimal scoring and the discriminant analysis problems are equivalent to the mixed functional-multivariate CCA problem of maximizing the covariance of $\int \xi x_i$ and $\boldsymbol{\eta}'\mathbf{y}_i$ subject to the constraints

$$\mathrm{var}(\int \xi x_i) + \lambda \mathrm{PEN}(\xi) = \mathrm{var}(\boldsymbol{\eta}'\mathbf{y}_i) = 1.$$

- This equivalence is because $\mathrm{Cov}(\{a_1, \ldots, a_n\}, \{b_1, \ldots, b_n\}) = (n-1)^{-1}(\sum_{i=1}^n a_i b_i - n\bar{a}\bar{b})$ and $\sum_{i=1}^n (a_i - b_i)^2 = \sum_{i=1}^n a_i^2 + \sum_{i=1}^n b_i^2 - 2\sum_{i=1}^n a_i b_i$.

Applications

Two applications were presented in Hastie et al. (1995):

- Speech recognition: frequency spectra of digitized recordings of various phonemes are used as data.
- Handwriting recognition: recognition of digits in handwritten postal addresses and zip codes, x_i are functions of a bivariate argument t, defined in practice on a 16×16 pixel grid.

Functional canonical correlation analysis

- Canonical correlation allows us to simplify the study of the ways in which two sets of functions are correlated.
- But we need to impose smoothness on the canonical weighting functions in order to keep the number of modes of variation with high correlations small.
- Its functional-vector version is connected with functional discriminant analysis.